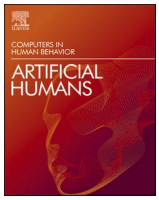
Computers in Human Behavior: Artificial Humans 8 (2026) 100302



Contents lists available at ScienceDirect

Computers in Human Behavior: Artificial Humans

journal homepage: www.journals.elsevier.com/computers-in-human-behavior-artificial-humans





Who performs better? Exploring the role of generative AI in creativity and brainstorming

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ARTICLE INFO

Keywords:
Creativity
Generative AI
Group brainstorming
Human-AI collaboration
Idea generation

ABSTRACT

Creativity is a fundamental human capacity, and while generative artificial intelligence (GenAI) is increasingly used to support creative work, the extent to which it can improve human creativity remains unclear. Innovation begins with the generation of creative ideas, and hybrid brainstorming (individual followed by team brainstorming) remains one of the most widely adopted technique for fostering creative ideas. Understanding how GenAI shapes individual idea generation is increasingly important, particularly when individuals anticipate future collaboration with either a human or an AI teammate. This study examines how incorporating GenAI as an assisting tool during individual brainstorming influences idea quantity, quality, and flexibility, and whether these effects vary by anticipated collaboration partner. In a laboratory experiment (N = 212), participants brainstormed either independently or with AI assistance while anticipating a subsequent collaboration with a human or an AI teammate. Results showed that individuals brainstorming independently generated a greater number and flexibility of ideas, whereas those who used AI assistance produced ideas of higher quality. Among AI-assisted participants, those anticipating a human teammate generated ideas that were more original and enjoyable than those anticipating an AI collaborator. These findings indicate that GenAI can enhance idea quality during individual brainstorming, but its benefits are shaped by social-motivational factors. Integrating GenAI into creative processes may be most effective when AI complements, rather than substitutes for, active human collaboration and engagement.

1. Introduction

Tech leaders have argued that human creativity and the generation of truly original ideas (Times of India, 2025) is a trait that will remain difficult for algorithms to replicate. Innovation begins with the generation of creative ideas, and group brainstorming remains one of the most widely adopted techniques for fostering creative ideas (Osborn, 1953). In particular hybrid-brainstorming, where participants first engage in individual idea generation (ideation) before collaborating with a teammate(s), is an effective approach for enhancing creativity and performance outcomes (Ahn et al., 2023; Rietzschel et al., 2010). This approach has high ecological validity as individuals often prepare ideas before engaging in a group meeting and discussion, making individual-then-group sequence ecologically valid and realistic (Ahn et al., 2023).

With the rise of generative artificial intelligence (GenAI) tools, the question of how AI influences human creativity has become increasingly relevant. As AI can be integrated across diverse domains, it opens new possibilities for both innovation and efficacy. The use of AI technologies can enable individuals to push creative boundaries by exploring novel ideas and combinations that might otherwise be overlooked (Agboola, 2024). In organizations, over half of employees report frequent AI use at work and even outside the workplace (McKinsey & Company, 2023; Vuong, 2026), suggesting a trend of increasing adaptation and incorporation of GenAI in the workplace. GenAI uses large language models, which can be trained on processing vast amounts of information, to generate human-like text, answer questions, and perform other language-related tasks with high accuracy (Kasneci et al., 2023; Routley, 2023). For example, in design and architecture, AI technologies can assist designers in generating solutions that enhance urban livability and resource management by analyzing large datasets (Agboola, 2024). Because GenAI can produce diverse ideas, refine text, and even suggest solutions on a given task, there can be potential benefits of using GenAI to support the process of idea generation in brainstorming.

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https://doi.org/10.1016/j.chbah.2026.100302
Received 15 February 2026; Received in revised form 25 March 2026; Accepted 25 March 2026
Available online 30 March 2026
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Within the framework of hybrid brainstorming, individuals can engage in both individual and collaborative stages of ideation, with AI integrated at both points in the process. For example, individuals may now consult with AI to develop ideas before attending a collaborative group brainstorming sessions and then during the group session, treat GenAI as an active teammate. This indicates that integrating AI into the brainstorming process can happen at multiple stages, as Malone (2018) proposed that computers can serve various functional roles in collaborations such as tools, assistants, peers, or managers. Individuals may use AI before engaging in group brainstorming as an assisted tool to potentially help them expand the quality and quantity of initial ideas. When used as a tool, GenAI remains subordinate to human input by offering suggestions (Malone, 2018). As GenAI tools become more sophisticated, they can take on multiple team roles (Sun et al., 2025). Specifically, AI as an active conversational partner and teammate can move beyond suggestions to evaluation and collective decision-making. Incorporating AI as a teammate has led to a new form of teaming: Human-Autonomy Team (HAT). HAT involves one or more autonomous agents working together with a human on interdependent tasks to achieve a shared goal (O'Neill et al., 2023).

Research on creativity demonstrates that idea generation is shaped by both cognitive and motivational processes. From a cognitive perspective, cognitive stimulation suggests that exposure to novel input can spark, through spreading activation, additional association and idea categories and expand an individual's creative potential. Spreading activation theory predicts that memory retrieval occurs through the activation of concept nodes within a semantic network (Collions & Loftus, 1975). Often, activation starting from an initial concept or node expands to all semantically related nodes and continues to spread (Collions & Loftus, 1975; Paulus & Kenworthy, 2019). In other words, the activation of a single concept can generate different related ideas, as new, associated concepts become activated. Hence, incorporating GenAI during the initial ideation stage may therefore amplify these associative benefits of cognitive stimulation compared to brainstorming independently (Doshi & Hauser, 2024).

At the same time, social motivational forces described by the Collective Effort Model (CEM) highlight that individuals adjust their effort in collective tasks based on whether they believe their personal contributions meaningfully influence valued outcomes (Karau & Williams, 1993). Because anticipating a human versus an AI teammate may shape these expectations differently, this framework is especially crucial and is discussed in greater detail later. Essentially, social evaluative cues are typically more salient when individuals anticipate collaborating with another human but may operate differently when the anticipated partner is an AI system, which is perceived as highly competent and capable. Yet existing research has focused primarily on AI's direct influence on creative output, leaving open questions about how AI shapes individuals' motivation, effort allocation, and prior idea generation before group interaction begins, particularly when AI assistance and expectations about future teammates coexist.

Therefore, our theoretical rationale is based on cognitive stimulation and spreading activation theory, as well as the Collective Effort Model. As prior research lacks an understanding of how AI shapes both motivational and cognitive aspects of creativity and idea generation prior to collaboration, this study addresses this theoretical gap by examining how GenAI assistance interacts with anticipated teammate type to influence individual creativity during the individual ideation stage of a hybrid brainstorming task. Specifically, we examine the extent to which AI-assisted idea generation influences creative outcomes compared to independent thinking, and how these effects interact when individuals expect to collaborate with either a human or an AI partner (see Fig. 1). By investigating both cognitive and social motivational dynamics, this study advances existing theories of brainstorming and human-AI interaction by demonstrating how cognitive stimulation and motivational dynamics, as outlined in the Collective Effort Model, jointly shape individual idea generation performance. This research also offers valuable insights for organizations seeking to integrate AI in ways that strengthen creative performance and optimizing future collaborative outcomes.

2. Conceptual foundation

2.1. Understanding brainstorming process

Brainstorming, first popularized by Osborn (1953), is a structured method for generating creative ideas in groups by encouraging as many ideas as possible. Successful brainstorming typically contains the four key principles, which are “deferring judgement, encouraging wild ideas, needing the quantity of ideas, and building on each other's ideas” (Ahn et al., 2023; Osborn, 1953, p. 302). Since its introduction, brainstorming has remained one of the most widely recommended techniques for creative problem-solving in group contexts because it enhances motivation and morale while allowing teams to draw on each members' expertise (Korde & Paulus, 2017; Paulus et al., 2001). Creative idea generation or ideation is a cognitive process that involves retrieving relevant knowledge from long-term memory and the integration of knowledge in working memory (Paulus & Brown, 2007), and group brainstorming seeks to expand that knowledge base.

2.2. Incorporating GenAI into hybrid brainstorming

Researchers have found that interacting brainstorming groups often underperform compared to pooling the outputs of individuals working alone (Diehl & Stroebe, 1987). This reduced performance has been attributed to process loss due to coordination difficulties, like having to wait to speak; social anxiety leading to initial low contributions; and then the development of norms of low contribution rates (Diehl & Stroebe, 1987). However, scholars have examined how using hybrid brainstorming and alternating between individual and group brainstorming sessions can enhance brainstorming performance compared to only group brainstorming by minimizing factors like potential process loss (Korde & Paulus, 2017). When individuals work alone first and then brainstorm in a group interaction, they are able to bring their best ideas first to the group. Ahn et al. (2023) argues that contrary to low initial contribution rates in just group brainstorming, when group members have brainstormed individually before the group brainstorm session, they often pick their best ideas to present first in order to impress other group members. This can then lead to norms of high contribution rates and to the best ideas coming early in brainstorming (Ahn et al., 2023; Korde & Paulus, 2017).

For hybrid brainstorming, alternating between individual and group ideation has become a defining characteristic of effective brainstorming because it balances independent thinking and collective synergy. Further, in workplaces, an individual often has a chance to prepare ideas before engaging in a group discussion, making individual-then-group sequence more ecologically valid and realistic.

Extending this framework to contemporary creative environments, hybrid brainstorming can also involve the integration of AI into the ideation process. The relationship between AI and human creativity can be examined in different ways depending on functional role AI plays in the creative process. Building on the notion that computers can serve multiple roles in collaborations, such as tools, assistants, peers, or managers (Malone, 2018), AI's creative influence can be conceptualized as either augmentation or transformation (Agboola & Yassin, 2025). Augmentation occurs when AI functions as an assisting tool that enhances the human creative process by providing analytical and generative capabilities, whereas transformation occurs when AI operates as a collaborative partner, reshaping traditional notions of authorship and originality in creative work (Agboola & Yassin, 2025).

Therefore, this paper adopts the hybrid brainstorming structure, having participants first engage in individual ideation while anticipating brainstorming collaboration with a future teammate. During the individual ideation stage, participants either generate ideas independently

or use AI as an assisting tool to help the idea-generation process. While AI may provide analytical or generative support related to the task, participants remain in control of the ideation output and type out the ideas after receiving feedback from AI.

2.3. Brainstorming performance – idea quantity, quality, and flexibility

Creativity in brainstorming is evaluated on three key dimensions: quantity, quality, and flexibility. Quantity, or fluency, reflects an individual's ability to generate a large number of ideas within a given time frame and reflects the number of ideas generated, regardless of their quality (Guilford, 1950).

Idea quality is defined by both originality and feasibility (Ahn et al., 2023; Diehl & Stroebe, 1987; Rietzschel et al., 2010). Originality represents the novelty and uniqueness of ideas to the given task, a hallmark of creative behavior, while feasibility measures the level of ideas that can be practically implemented (Rietzschel et al., 2010; Runco & Charles, 1993). Therefore, for an idea to be considered truly creative and good, it must strike a balance between being novel and new and being useful in practice. Additionally, prior work has extended the idea quality metrics of originality and feasibility by incorporating task-relevant criteria such as usefulness, in the context of backpack design task (Ahn et al., 2023). In a similar vein, we include an additional evaluative dimension to capture the intrinsic appeal of an idea by assessing the extent to which ideas are perceived as fun or enjoyable. This measure of idea quality complements standard creativity indicators by capturing motivational aspect of ideation, which are theoretically relevant to individuals' engagement in the creative process. In addition, given that the task assigned to participants in this study is to help redesign a college orientation event, the concept of fun and enjoyable is task-relevant, which we provide more details about in the methods section.

Beyond the quantity and quality of ideas, flexibility, referring to the diversity or number of distinct idea categories, is also a crucial indicator of creativity. According to spreading activation theory when a category or node is activated in working memory, the activation spreads more readily to semantically related concepts that are closely connected to the activated node (Collions & Loftus, 1975). For example, thinking of the concept "orange" could activate the category of fruit but could also activate categories like color, Oompa Loompa, varieties of soda, etc., each of which would then activate related nodes: blue, Willie Wonka, grape. In other words, generating ideas across multiple conceptual categories represents that participants are accessing broader and more heterogeneous areas of their semantic networks. Therefore, an idea in a new category can be more valuable than an additional idea in a previously used category because it suggests a shift to a new cognitive domain rather than staying within the existing area (Nijstad & Stroebe, 2006).

3. Theoretical background

Individual idea generation is shaped by a variety of cognitive and motivational factors such as one's knowledge in the specific area and the motivation to make use of the knowledge. Three factors influence individuals' performance during individual idea generation, which are cognitive stimulation, social loafing, and impression management. Therefore, we next discuss the theoretical rationale behind these factors and the implications of using GenAI.

3.1. Cognitive stimulation

Cognitive stimulation, based on spreading activation and association, refers to the process where one person's idea triggers new ideas, creating a "chain reaction" of ideation (Osborn, 1953). GenAI can serve as a powerful catalyst for stimulating cognitive connections and mental associations for creative tasks (Delul et al., 2021; Doshi & Hauser, 2024). Research on creative story writing has shown that "generated ideas from AI may be used as a 'springboard' for the human mind, providing potential starting points that can result in a 'tree structure' of different storyline" (Doshi & Hauser, 2024, p. 1). In other words, GenAI has the capacity to help individuals develop creative ideas and enhance new potential associations for creative idea generation.

Further evidence comes from an applied organizational context. A study conducted at the Boston Consulting Group found that consultants who used GenAI performed significantly better than those working independently on innovative multi-stage tasks such as product development and marketing strategy (Dell'Acqua et al., 2023). Consultants assisted by AI produced higher quality results and progressed through tasks more efficiently. These findings indicate that by offering rapid feedback and having alternative perspectives on cognitive tasks, AI tools can spark new associations and even reduce the cognitive load, ultimately enhancing creative performance.

Creativity research has long documented a positive relationship between idea quantity and quality, meaning the more ideas individuals generate, the greater the likelihood of producing high quality ideas (Diehl & Stroebe, 1987; Dugosh & Paulus, 2005; Rietzschel et al., 2010). In other words, generating a larger pool of ideas increases the probability of arriving at more original and innovative solutions. Beyond the quantity and quality, idea flexibility, meaning the diversity of idea categories, is another essential dimension of creativity, because even a small increase in the number of distinct categories can substantially enhance creativity (Ahn et al., 2023).

Although GenAI appears to enhance idea quality and facilitate cognitive stimulation, evidence also suggests that it may simultaneously constrain idea flexibility. In other words, there was a notable lack of idea variability for participants with GenAI assistance compared to those working independently (Dell'Acqua et al., 2023). A similar pattern emerged in research where participants used GenAI for brainstorming and writing short stories. Doshi and Hauser (2024) found that although stories created with the assistance of AI tended to be more engaging, appropriate, feasible, and even publishable, with more access to AI, there is an increase in similarity among composed stories. These patterns suggest that while AI can be beneficial in individual creativity, it may also constrain overall idea variability by yielding similar results. Based on this research, we pre-registered this hypothesis:

Hypothesis 1. Individuals who received AI assistance will produce ideas of higher quantity and quality, but lower flexibility than individuals working alone.

3.2. The Collective Effort Model (CEM)

Extensive research has shown that people often exert less effort when working collectively than working alone (Karau & Williams, 2001). This tendency known as social loafing has been found across a wide range of group tasks. Classic examples include the rope-pulling studies, where participants in larger groups exerted substantially less force than would be expected based on the sum of individual effort (see Kravitz & Martin, 1986). Similar patterns have been replicated in cognitive tasks such as idea generation (Karau & Williams, 2001).

The Collective Effort Model (CEM) provides a strong theoretical account of why such reduced effort occurs when working with others or anticipating working with others. CEM suggests that individuals are motivated to exert effort on collective tasks only when they expect their contributions are instrumental or essential in obtaining outcomes that they value personally (Karau & Williams, 2001). The model suggests that working on a collective task introduces additional motivational barriers based on whether one's contributions affect group performance and translated into an outcome that is personally valued, such as an objective higher reward or the subjective outcome of being evaluated more positively by others.

Importantly, CEM emphasizes that motivation increases when tasks provide clear information about one's individual performance, especially when others can observe or even evaluate that performance.

Research consistently shows that individuals are motivated to maintain a favorable self-evaluation and to present themselves positively in group contexts (see Ahn et al., 2023; Brown et al., 1988; Sutton & Hargadon, 1996). This motivational tendency directly aligns with the theory of impression management, which argues that individuals consciously regulate or pay attention to their behavior to create and maintain positive perceptions in the eyes of others (Schneider, 1981). However, this line of research has primarily been conducted in humans' interaction contexts. Because people often assume that AI does not judge them, impression management concerns should be lower when individuals anticipate collaborating with an AI teammate (Diwanji et al., 2025; Lucas et al., 2014). This difference has important implications for whether people reduce effort or engage in social loafing when expecting to work with an AI teammate.

Recent brainstorming research further supports this tendency of impression management. When individuals brainstorm individually before group interactions, they tend to subsequently share their best ideas first with the group due to impression management (Ahn et al., 2023). More broadly, Baumeister and Leary (1995) argued that the need to belong and to establish and maintain strong interpersonal bonds is one of the most fundamental human motivations. Therefore, when participants anticipate working with a human partner, especially the presence of the other person in the room, can heighten their motivation to perform well, exert more effort, and potentially make meaningful contributions for the subsequent session. We pre-registered this hypothesis:

Hypothesis 2. Individuals anticipating a human teammate will produce ideas of higher quantity, quality, and flexibility compared to those anticipating an AI teammate.

Moreover, a meta-analysis of 78 studies by Karau and Williams (1993) found that expectations about teammate performance are a significant predictor of the tendency for one to social loaf when engaging in collective task compared to working individually. Specifically, individuals were more likely to loaf when they expected their partners to perform well, but exerted greater effort when they anticipated that their partners would perform poorly. Thus, if individuals feel that their contributions may not affect the group outcome much in comparison to their partner, they social loaf. Researchers have supported that people tend to have an expectation of high performance from AI (Madhavan & Wiegmann, 2007; Prahl & van Swol, 2021). Whether individuals expect to collaborate with an AI partner, perceiving the AI as highly capable may lead them to rely on its input and reduce their own effort, assuming their own input may not make a meaningful contribution beyond the AI.

This tendency aligns with research on algorithmic appreciation. Although algorithmic appreciation has primarily been examined in advice utilization contexts, it captures a broader tendency for individuals to place greater weight on information when it is labeled as coming from an algorithmic versus human source (Logg et al., 2019). Logg and colleagues (2019) demonstrated that people not only valued but also relied more heavily on information they believed was generated algorithmically. Thus, when people have algorithmic appreciation and have high expectations for AI performance, they engage in "algorithmic loafing" (Inuwa-Dutse et al., 2023).

According to the Collective Effort Model, individuals reduce their effort when a group member is perceived as highly competent or more capable of driving group performance. Thus, participants who anticipate collaborating with an AI teammate, whom might be viewed as highly competent, may decrease their effort even before engaging in the group interaction by assuming that their AI partner will do a better job. Taken together, these mechanisms imply that the impact of AI assistance during ideation will depend on whom participants believe they will collaborate with next. Building on the first two hypotheses, which separately addressed the effects of AI assistance and anticipated teammate on ideation outcomes, we predict an interaction effect between the ideation condition and the anticipated teammate. We have these non-preregistered hypotheses that:

Hypothesis 3a. Idea quantity and quality will be highest when individuals receive AI assistance and anticipate a human teammate (AI-HHT), followed by no-AI assistance and anticipate a human teammate (HHT), followed by no-AI assistance with an anticipated AI teammate (HAT), and lowest when individuals receive AI assistance with an anticipated AI teammate (AI-HAT).

Hypothesis 3b. Idea flexibility will be highest when individuals do not receive AI assistance and anticipate a human teammate (HHT), followed by AI-assisted ideation with an anticipated human teammate (AI-HHT), followed by no-AI assistance with an anticipated AI teammate (HAT), and lowest when individuals receive AI assistance with an anticipated AI teammate (AI-HAT).

4. Method

The study employed a 2 × 2 between-subjects design manipulating ideation condition (independent vs. AI-assisted) and anticipated teammate (human vs. AI; see Fig. 1). A priori power analysis indicated that a sample of $N = 128$ would be sufficient to detect a small-to-medium effect size ($f = .25$) with 80% power. Although the distribution of participants across conditions was uneven due to the study being nested within a larger project, the final sample of $N = 212$ participants yield an estimated post-hoc power of 95%, indicating adequate sensitivity to detect effects of the targeted magnitude. Participants were randomly assigned to one of the four conditions (Independent/Anticipated human teammate = 69; AI-assisted/Anticipated human teammate = 68; Independent/Anticipated AI teammate = 37; AI-assisted/Anticipated AI teammate = 38).

4.1. Participants

This study received approval from the Institutional Review Board (IRB). All participants provided informed consent prior to participation and were acknowledged that their responses would remain confidential and used only for research purposes. Participants were also informed that they could withdraw from the study at any time. A total of 224 participants were recruited for this in-person lab experiment at a large public university in the Midwestern United States. Ten participants failed the condition-check items assessing whether they correctly identified their anticipated partner or if they received the AI assistance and were, therefore, excluded from the analyses. Two participants were also excluded from the study: one due to a facilitator error in failing to provide the required written handout, and the other due to a language barrier that made it difficult for the participant to clearly articulate her ideas in writing. The final sample consisted of 212 undergraduate students (75.5% female, 23.6% male, .9% other). Participants' average age was 20.02 ($SD = 1.67$); 69.3% White, 4.2% Black, 23.1% Asian, 1.9% American Indian; .5% Pacific Islander, and 3.8% preferred not to answer. Participation was voluntary, and students received one extra course credit for their participation. In addition, participants could receive a $20 gift card if they were in the top 25% of task performance.

4.2. Study design

Participants were asked to complete a 10-min context specific brainstorming task aimed at generating ideas to improve the university's *Student Orientation, Advising, and Registration* (SOAR) program. In the independent ideation condition, participants brainstormed alone. In the AI-assisted condition, participants used Microsoft Co-pilot (Auto mode), a university-licensed generative AI system, to support their idea generation. Microsoft Co-pilot was selected for two primary reasons. First, as a university-licensed platform, it meets institutional data-security and privacy standards, ensuring the protection and confidentially of participants' responses. Second, each participant accessed the system through

an individual account, which reduced shared-account use and minimized the potential for cross-participant data contamination or AI self-learning effects that could arise from repeated interactions through a single account. Before beginning the task, participants received standardized instructions explaining that Co-Pilot should be treated as an assisting tool for asking specific questions or getting suggestions, rather than using AI-generated responses verbatim. Participants had to type out their ideas and could not use AI-generated responses verbatim.

To manipulate the anticipated teammate, participants were informed about the nature of their future collaborator for a subsequent group brainstorming session. They were told that their future teammate would be either a human partner or an AI teammate. This design allows the study to further explore how anticipating collaboration with a human versus an AI partner influences individuals' initial stage of idea generation performance. Participants who anticipated a human teammate were paired randomly according to the sign-up schedule. When two participants arrived for the same session, they were paired with each other, and the presence of one another human served as a cue for their future teammate. Participants who arrived alone were assigned to the condition anticipating an AI teammate. Paired participants were strangers and did not know each other prior to the study. After the completion of the brainstorming task, participants completed a post-study questionnaire that included a manipulation check of impression management and measurements regarding one's motivation to engage in the task.

4.3. Procedure

Upon arrival, participants completed an informed consent form. The researcher provided an overview of the study's purpose and procedure and informed participants about the nature of their anticipated partner for the potential subsequent group brainstorming session. No deception was used.

After obtaining informed consent, participants watched a brief 2-min video demonstrating traditional group brainstorming techniques developed by IDEO. The researcher then emphasized the four core rules of brainstorming, which are deferring judgment, encouraging wild ideas, needing the quantity of ideas, and building on each other's ideas (Osborn, 1953), and provided a printed copy for reference throughout the session. A pilot test using an 8-min individual ideation session on the same task revealed that the duration was insufficient, particularly in the AI-assisted condition, as participants needed additional time to type and wait for the responses. Hence, the individual idea generation phase was extended to 10 min in the experiment. After the video, participants were then presented with the main brainstorming task:

> "Student Orientation, Advising, and Registration (SOAR) is a specific program designed to transition incoming students as a new "Badger". You are a member of SOAR planning team! Recently, however, SOAR has received several complaints from students that SOAR is not fun, so your job is to improve or redesign this program to make it more engaging and helpful for future students. The goal is to ensure that the future Badgers have a smooth and enjoyable transition into their university life."

Participants in the AI-assisted condition were instructed to use Microsoft Co-Pilot as an assisting tool by asking "what" and "how" questions to generate ideas. Each participant logged into their university-licensed Co-Pilot account, and researchers provided step-by-step instructions. The research team assured participants that all interactions would remain confidential and that AI chat histories would be deleted immediately after data collection to ensure privacy.

4.4. Measures

Quantity of ideas. The total number of unique ideas typed by participants in the Word document was taken as quantity of ideas.

Quality of ideas. Trained undergraduate coders, blind to condition, used a coding scheme to rate ideas. Ideas were assessed on three dimensions: 1). originality, representing the novelty and uniqueness of the idea; 2). feasibility, referring to the ease of implementation of the idea; 3). fun, representing the perceived enjoyability of the idea. Each coder rated each dimension on a five-point scale ranging from 1 (*poor*) to 5 (*excellent*). A mean score was calculated for each participant by averaging their ratings across all generated ideas. Because data were collected in two semesters, coding was conducted in two corresponding periods. During the first period, four trained coders jointly coded approximately 16% of the ideas to establish interrater reliability. The interclass correlation coefficient (ICCs) were .76 for originality, .70 for feasibility, and .71 for fun, all exceeding the commonly accepted .70 threshold, indicating strong reliability across the three quality dimensions. During the second coding period, one coder was replaced due to scheduling constraints. To re-establish reliability, the new coder and the continuing coder independently coded approximately 20% of the ideas, yielding acceptable intercoder reliability with ICCs = .84, .78, and .75 respectively for originality, feasibility, and fun. Additionally, the averaged ratings from those jointly coded subsets served as the final scores for those ideas. After reliability was confirmed for each period, the remaining ideas were randomly distributed among the coders, who independently rated them using a blind-coding procedure.

Flexibility of ideas. Idea flexibility was assessed by counting the number of distinct idea categories produced by each participant, reflecting the diversity of their idea pool. The categorization scheme was developed collaboratively through an in-depth discussion involving the research team and trained coders. The team first generated preliminary notes on recurring idea types, then through discussion, refined these observations into a comprehensive set of thirteen mutually exclusive categories that captured the full range of responses without imposing overly narrow distinctions. For example, although activities such as going on a picnic, playing bingo, participating in icebreaker games, or hosting a flea market represent different specific suggestions, they were all classified under the broader category of social activities. After the coding scheme was finalized, trained coders jointly coded ideas from 15 participants to establish consistency, so any discrepancies were discussed and resolved through consensus. Following this in-depth discussion, the coders were randomly assigned an equivalent number of participants and independently coded participants' ideas according to the established coding scheme, while remaining blind to the experimental condition. The flexibility score for each participant was calculated as the total number of unique categories represented in their idea list.

Impression management. Participants' perception of their impression management during the task was measured using eleven items adapted from Wayne and Ferris (1990). All items were rated on a 7-point Likert scale (from 1 = *strongly disagree* to 7 = *strongly agree*). Because participants anticipated to interact with either a human or an AI partner, item wording was adjusted slightly by replacing "partner" with "AI partner" in the AI condition. Although eleven items were included in the full questionnaire, six of them focused specifically on behaviors occurring during or after the interaction with the partner. Because the present study aimed to assess impression management during the individual idea generation phase, five items most appropriate for capturing pre-interaction impression management were used for this study. These items include, "do you present your partner as a friendly and approachable person?", "do you let your partner know you are committed to doing a good job?", "do you work harder because you will be evaluated by your partner?", "do you make your partner aware of your contribution to the team's effort?", and "do you make sure your partner sees you as responsible for team's positive outcomes?". Responses on these five items were averaged to present participants' impression management score during the individual idea generation phase (Cronbach's alpha = .75).

Demographics. Participants reported their age, gender, ethnicity,

race, and academic major. Residency status was also collected to ensure accurate distribution of monetary rewards.

4.5. Data analysis

The present study employs quantitative analytical methods to examine differences in creative performance across experimental conditions. Participants' responses were collected in individual documents and subsequently coded to generate quantitative indicators of idea performance. Creative performance was operationalized along multiple dimensions, including idea quantity, quality, and flexibility. Trained coders evaluated all responses using predefined coding criteria. Inter-coder reliability was assessed using intraclass correlation coefficients (ICC), indicating acceptable levels of agreement. Idea flexibility was assessed through a categorization approach based on a predefined coding scheme developed through iterative coder discussion. To test the hypotheses, analyses of variance (ANOVA) were conducted. Where significant interaction effects were identified, follow-up simple effects were examined using pairwise comparisons of estimated marginal means with Bonferroni adjustment. All statistical analyses were performed using SPSS software version 27.

5. Results

Unless otherwise stated, results are analyzed in a 2 (Independent vs. AI-assisted) by 2 (Anticipated Human vs. AI teammate) ANOVA. Descriptive statistics are presented in Table 1 and correlations are shown in Table 2.

5.1. Impression management manipulation check

The main effect of ideation condition on participants' impression management was not significant, $F(1, 208) = .413$, $\eta^2 = .002$, $p = .521$, nor was the interaction between the two factors, $F(1, 208) = .662$, $\eta^2 = .003$, $p = .417$. In contrast, there was a significant main effect of anticipated teammate, $F(1, 208) = 93.85$, $\eta^2 = .311$, $p < .001$, such that participants who anticipated collaborating with a human partner reported higher levels of impression management concerns than those anticipating an AI partner. The significant effect of anticipated teammate provides evidence that the anticipated teammate manipulation successfully elicited differential evaluative concerns when participants expected a human, rather than an AI, teammate.

5.2. Quantity of ideas

The main effect of ideation condition was significant, $F(1, 208) = 4.14$, $\eta^2 = .020$, $p = .043$, indicating that participants who generated ideas independently produced significantly more ideas than those using AI assistance. However, the main effect of anticipated teammate was not significant $F(1, 208) = .010$, $\eta^2 = .000$, $p = .919$, nor was the interaction between the two factors, $F(1, 208) = 1.53$, $\eta^2 = .007$, $p = .217$. Contrary to our pre-registered Hypothesis 1, AI assistance was associated with lower idea quantity compared to independent brainstorming.

5.3. Quality of idea

Originality. The main effect of ideation condition on originality of ideas generated was significant, $F(1, 208) = 16.78$, $\eta^2 = .075$, $p < .001$, indicating that participants who received AI assistance generated more original ideas than those who worked independently. This supports the quality component of our pre-registered Hypothesis 1, specifically with respect to idea originality. Additionally, the main effect of anticipated teammate $F(1, 208) = 7.47$, $\eta^2 = .035$, $p = .007$ was significant. This suggests that participants anticipating a human partner outperformed those anticipating collaboration with an AI teammate, supporting our pre-registered Hypothesis 2 that anticipating a human partner led to higher idea quality.

Additionally, the interaction between ideation condition and the anticipated teammate was significant, $F(1, 208) = 7.61$, $\eta^2 = .035$, $p = .006$. Partially consistent with Hypothesis 3a, the exploratory examination of the interaction between ideation condition and anticipated teammate revealed a differentiated pattern for idea originality. Participants who received AI assistance while anticipating a human partner generated the most original ideas. Bonferroni-adjusted simple effects comparisons showed that, when participants anticipated a human teammate, there was a significant difference in originality for AI-assisted ideation with substantially higher originality (M = 3.49, SD = .59) compared to independent ideation (M = 2.99, SD = .47), $F(1, 135) = 29.94$, $\eta^2 = .182$; $p < .001$, 95% CI [.32, .68]. However, when participants anticipated an AI teammate, there was no significant difference in levels of originality for those who received the AI assistance (M = 3.09, SD = .43) compared to those who worked independently (M = 2.99, SD = .48), $F(1, 73) = .87$, $\eta^2 = .012$; $p = .36$.

Feasibility. The main effect of ideation condition on feasibility of ideas generated was not significant, $F(1, 208) = 2.71$, $\eta^2 = .013$, $p = .101$. Thus, this result does not provide support for the feasibility component of the quality prediction in Hypothesis 1. The main effect of anticipated teammate $F(1, 208) = 1.67$, $\eta^2 = .008$, $p = .197$ was also not significant, which also fail to provide support for the feasibility component of the quality prediction in Hypothesis 2.

However, there was a significant interaction effect between the ideation condition and anticipated teammate, $F(1, 208) = 10.80$, $\eta^2 = .049$, $p = .001$. Inconsistent with Hypothesis 3a, the exploratory examination of the interaction between ideation condition and anticipated teammate revealed a differentiated pattern for idea feasibility. Participants who brainstormed independently while anticipating an AI partner generated the most feasible ideas. Bonferroni-adjusted simple effects comparisons showed that, when participants anticipated a human teammate, there was no significant difference in levels of feasibility for those who received the AI assistance (M = 3.51, SD = .66) compared to those who worked independently (M = 3.37, SD = .59), $F(1, 135) = 1.85$, $\eta^2 = .014$; $p = .176$. In contrast, when participants anticipated an AI teammate, AI-assisted ideation resulted in significantly lower feasibility (M = 3.10, SD = .70) compared to independent ideation (M = 3.54, SD = .49), $F(1, 73) = 9.94$, $\eta^2 = .120$; $p = .002$, 95% CI [−.72, −.16].

Fun. The main effect of ideation condition on overall fun and enjoyableness of ideas generated was significant, $F(1, 208) = 4.23$, $\eta^2 = .020$, $p = .041$, indicating that participants who received AI assistance generated ideas that are more fun than those worked independently. This supports the quality component of our pre-registered Hypothesis 1, especially the enjoyableness of ideas. Additionally, the main effect of anticipated teammate $F(1, 208) = 20.07$, $\eta^2 = .088$, $p < .001$ was significant, suggesting those who anticipating a human partner produced ideas that are more interesting and enjoyable compared to those anticipating an AI teammate. This supports our pre-registered Hypothesis 2 that anticipating a human partner led to higher idea quality.

Moreover, a significant interaction emerged between ideation condition and anticipated teammate, $F(1, 208) = 9.58$, $\eta^2 = .044$, $p = .002$. Exploratory examination of this interaction revealed a pattern for idea enjoyableness that was consistent with Hypothesis 3a that participants who received AI assistance while anticipating a human partner produced ideas rated as the most enjoyable. Bonferroni-adjusted simple effects comparisons showed that, when participants anticipated a human teammate, there was a significant difference in idea enjoyableness for AI-assisted ideation which led to substantially higher score (M = 3.68, SD = .69) compared to independent ideation (M = 3.23, SD = .58), $F(1, 135) = 17.25$, $\eta^2 = .113$; $p < .001$, 95% CI [.24, .67]. However, when participants anticipated an AI teammate, there was no significant difference in levels of fun for those who received the AI assistance (M = 3.01, SD = .59) compared to those who worked independently (M = 3.10, SD = .52), $F(1, 73) = .50$, $\eta^2 = .007$; $p = .48$.

5.4. Flexibility of idea

The main effect of ideation condition on the flexibility of ideas generated was significant, $F(1, 208) = 5.05$, $\eta^2 = .024$, $p = .026$, indicating that participants who generated ideas independently produced greater variability in idea categories compared to those who received AI assistance. This supports the flexibility component our pre-registered Hypothesis 1. However, the main effect of anticipated teammate $F(1, 208) = .442$, $\eta^2 = .002$, $p = .517$, and the interaction between the two factors, $F(1, 208) = .65$, $\eta^2 = .003$, $p = .421$, were not significant. Accordingly, these results do not provide support for the flexibility component of Hypothesis 2, nor for Hypothesis 3b.

6. Discussion

This study investigated how incorporating AI during individual ideation influences creativity and how these effects depend on the social context of different types of anticipated teammate. Specifically, we examined whether individuals who receive AI assistance would generate ideas of higher quantity, quality, but lower flexibility than those working independently, and whether individuals anticipating an AI teammate would produce ideas of lower quantity, quality, and flexibility than those anticipating a human teammate. Lastly, we tested the potential interaction effects between ideation condition and anticipated teammate.

6.1. Influence of AI assistance

Regarding the effects of receiving AI assistance during individual ideation, the results partially supported our hypothesis. Participants who received AI assistance produced ideas that were significantly more original and enjoyable than those who worked independently, yet their ideas exhibited lower flexibility (in support of pre-registered Hypothesis 1). This pattern aligns with prior works demonstrating AI's capacity to enhance cognitive stimulation and deepen individuals' exploration within a particular conceptual space, thereby improving the richness and refinement of ideas (see Dell'Acqua et al., 2023; Doshi & Hauser, 2024).

The fact that participants generated more flexibility or the categories of ideas when brainstorming independently than when using GenAI can be explained through the lens of the "jagged frontier" of AI performance. The "jagged frontiers" metaphor characterizes the uneven landscape of AI capabilities across different tasks, suggesting that GenAI will perform better than humans if the tasks are within their learned patterns, but when tasks lie outside its programmed capacity, humans perform better without use of GenAI (Dell'Acqua et al., 2023; Martines, 2023). Idea flexibility, which reflects one's ability to expand beyond the previous conceptual node or shifting to a different perspective, is precisely the type of outcome that lies outside of the frontier. When participants received assistance from GenAI, it has the tendency to draw results from the same conceptual domain, leading to an increased chance of anchored thinking and reducing the overall idea diversity. In contrast, independent ideation may have allowed participants to engage in less constrained associative thinking, resulting in greater flexibility across ideas. Our finding is aligned with previous findings that when making complex decisions or engaging in more nuances thinking that require the integration of diverse information and exploratory recombination, using GenAI often underperforms or even makes simple mistakes compared to working alone (Dell'Acqua et al., 2023; Martines, 2023). This study provides evidence that humans have a cognitive advantage in brainstorming when it comes to idea diversity, highlighting a potential limitation of GenAI in supporting creative ideation.

Contrary to our hypothesis 1, participants brainstorming independently generated a significantly higher number of ideas. This finding on quantity of ideas suggests that even though AI may help individuals elaborate and polish ideas, the guidance it provides can constrain divergent thinking within a given timeframe by narrowing attention toward algorithmically suggested directions and fixating on a particular idea.

Taken together, these findings suggest a meaningful trade-off between breadth and depth in the ideation process. While AI assistance appears to help participants generate ideas that are more original or enjoyable, independent brainstorming appears to better support the production of larger number of ideas and a wider range of idea categories. Possibly, AI initially presented the participant with a few ideas, and in line with the Collective Effort Model, this may have provided a low norm of idea generation which participants followed (Ahn et al., 2023; Brown, Collions & Schmidt, 1998). This may have anchored participants' thinking within a narrower conceptual space and reduced their motivation to engage in broader divergent exploration.

In addition, there may have been process loss involved in working with AI, as participants had to spend time reading, interpreting, and evaluating AI-generated suggestions rather than generating ideas continuously on their own. In creative tasks where free exploration is essential, this form of refinement may function as a cognitive constraint, limiting both the number and categories of ideas individuals are able to generate. This also suggests a potential three-stage model of brainstorming that may be ideal: individuals first generate ideas independently, then expand or refine them with AI assistance, before engaging in collaborative discussion with human teammate.

6.2. Influence of anticipated teammate

Additionally, our results partially supported the predictions regarding anticipated teammate type. As expected, anticipating a human teammate led participants to generate ideas that were significantly more original and more enjoyable (in support of Hypothesis 2). However, idea quantity, feasibility, and flexibility did not differ across teammate conditions. The selective reduction in perceived novelty and enjoyableness of ideas when participants anticipated an AI rather than a human teammate may reflect social loafing, arising from the two factors of reduced social presence with its lower impression management concerns and the perception that GenAI could serve as a sufficiently competent partner, in support of the Collective Effort Model (Karau & Williams, 2001; Madhavan & Wiegmann, 2007; Prahl & van Swol, 2021).

In contrast, knowing they would later share their ideas with a human partner likely increased participants' social awareness and motivated them to refine their ideas in ways that would create a favorable impression. Our manipulation checks confirmed that participants had higher impression management concerns when they anticipated interacting with a human in the group brainstorming phase of the experiment. Prior research shows that individuals tend to bring up their best ideas early in group settings because people seek to present themselves positively to teammates (Ahn et al., 2023). Anticipating a human teammate appears to selectively enhance specific dimensions of creativity that are especially socially evaluative, such as originality and enjoyment, without broadly increasing effort to generate more ideas or expanding idea diversity. Possibly, participants did not anticipate as much social approbation from introducing feasible ideas. Participants may not be striving to present their human partner with a lot of ideas or diverse category of ideas, but rather more original and fun ideas than if their partner was AI.

6.3. Interaction effects of AI assistance and anticipated teammate

Most importantly, we observed a significant interaction between ideation condition and anticipated teammate. Participants who received AI assistance and anticipated working with a human partner produced ideas that were highest in originality and enjoyment compared to all other conditions. Together, these patterns highlight the powerful role of social motivational cues of anticipating a human collaborator,

prompting individuals to produce more original and enjoyable ideas they expect to share and discuss. At the same time, AI assistance enhances idea quality such as novelty and enjoyableness, but these benefits are most salient when individuals remain socially engaged and have higher impression management concerns due to an upcoming human partner. Consistent with this interpretation, simple effect analyses showed that when participants anticipated a human teammate, AI-assisted ideation led to higher levels of originality and enjoyment compared to independent ideation.

To further understand the role of social-motivational factor, we examined participants' impression management concerns. As expected, participants who anticipated collaborating with a human partner reported significantly higher impression management compared to those anticipating an AI teammate. This pattern aligns with the Collective Effort Model, which posits that individuals exert greater effort when their individual contributions are identifiable and subject to observation or evaluation by others (Karau & Williams, 2001). In other words, anticipating a human partner may have heightened concerns about self-presentation and evaluation, thereby increasing motivation during idea generation.

6.4. Theoretical contributions and ethical concerns

Overall, this study contributes to the human-AI collaboration by moving beyond the question of whether GenAI improves creative outcomes; instead, it examines when and under what social-motivational conditions its benefit emerge. In particular, the theoretical gap addressed in the study is how GenAI shapes idea generation before collaboration begins, particularly when cognitive support from AI and social-motivational expectations about a future teammate operate at the same time. This study shows that the effects of GenAI are not uniform across creativity dimensions and are socially contingent. Our findings suggest that GenAI is most creatively beneficial when embedded within workflows that preserve human to human collaboration, rather than replacing it. The results highlight that GenAI may not fully motivate or expand creative thinking on its own; instead, its advantages are amplified only when paired with the social presence, accountability, and evaluative expectations associated with human teammates.

Additionally, these findings also speak to emerging ethical concerns surrounding AI-assisted work. Scholars have argued that although automation in creative workflows has become a powerful tool to enhance efficiency and boost productivity, the question of who owns the rights to these creations remains complex (Agboola, 2024). Moreover, machine learning systems, despite growing sophistication, remain constrained by issues such as bias, fairness, environmental sustainability, explainability, and the limits of data-driven pattern recognition particularly when tasks require nuanced, context-specific judgment rather than output alone (Barbierato et al., 2025). From these perspectives, the present study highlights that GenAI is less effective in supporting forms of creativity that require broad divergent exploration and contextual knowledge. In other words, AI may assist the generation of more polished ideas, but that does not necessarily mean it can fully reproduce the more flexible and situationally grounded aspects of human creativity.

For organizations, this implies that integrating AI into team processes should be done in ways that complement rather than substitute the social dynamics that drive creative engagement. Designing AI-supported brainstorming sessions that maintain strong human collaboration cues may have more innovative and socially meaningful outcomes.

7. Limitations and future research

However, there are several limitations that should be considered for the future research. First, although our theoretical framework drew on mechanisms such as social loafing and cognitive stimulation, we did not include direct measures of these constructs. As a result, our ability to empirically test their relative contribution is limited, so future research should incorporate validated measures to capture those more precisely.

Second, future research would benefit from longitudinal approaches that examine how repeated or sustained use of GenAI influences individuals' ability to engage in creative tasks over time. Additionally, the use of a student sample may limit the generalizability of the findings. Although participants were drawn from diverse academic backgrounds and engaged in a context-specific task designed to mimic real-world creative settings, their experiences may differ from those of individuals in organizational settings. In particular, workplace environment often involves more complex tasks, higher stakes, and stronger social-motivational pressure. Therefore, future studies should examine similar processes in applied settings, such as organizational or professional contexts, to better understand how GenAI shapes creativity when social-motivational concerns are more salient and consequential.

8. Conclusion

In conclusion, this study demonstrates that incorporating AI during individual ideation stage can enhance the originality and enjoyableness of ideas but may simultaneously constrain flexibility and quantity. As innovation still remains to be a fundamental human trait, our results note the advantage that humans have in exploring more categories of ideas, that could be especially important towards making creative connections across categories and moving beyond the frontier of what is known. The findings further suggest that integrating AI tools into team processes may enhance innovation only when individuals will be actively engaged with human collaborators rather than only relying on AI. In essence, these results highlight that creativity in human-AI contexts is not only about how people use technology but also about how they situate it within social expectations. At a broader societal level, the findings demonstrate that AI integration should not replace human collaboration but rather complement it.

CRediT authorship contribution statement

Ika Dai: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Lyn M. van Swol:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Methodology, Conceptualization.

Declaration of informed consent

Informed consent was obtained from all individual participants included in the study. Participation was voluntary and anonymous.

Exemption statement

This study was reviewed and approved by the Institutional Review Board of University of Wisconsin-Madison (Approval No. 2025-0106).

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This research was supported by the Marilynn R. Baxter Trust Gift. The authors would like to thank Maya M. Anderson, Charlie J. Keith, Melody Cui, and Alex Zhao for their assistance with data coding and their support throughout the study.

Appendix

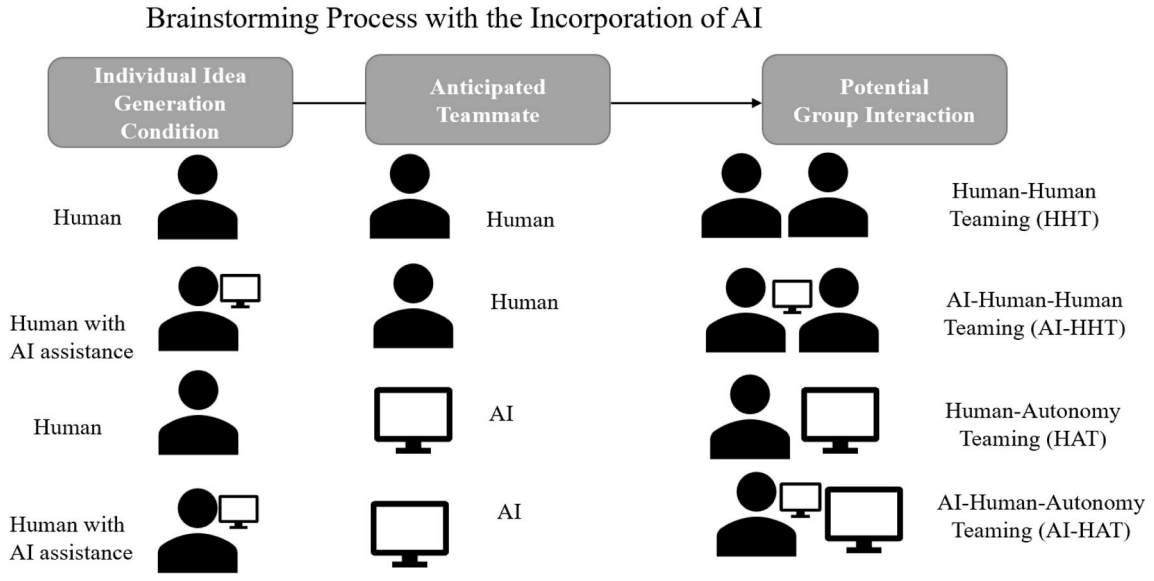


Fig. 1. Brainstorming Process with the Incorporation of AI

Table 1
Summary of Study Measurements in Descriptive Statistics

DV	IV							
	Ideation Condition				Anticipated Teammate			
	Independent		AI-Assisted		Human		AI	
	Mean	*SD*	Mean	*SD*	Mean	*SD*	Mean	*SD*
Quantity	13.24*	5.8	11.69*	7.00	12.50	6.80	12.39	6.46
Quality – Originality	2.99***	.47	3.34***	.56	3.24**	.59	3.04**	.45
Quality – Feasibility	3.43	.56	3.37	.70	3.44	.63	3.32	.64
Quality – Fun	3.18*	.56	3.44*	.73	3.45***	.67	3.06***	.56
Flexibility	6.66*	1.95	6.06*	2.22	6.29	2.17	6.48	2.00
Impression Management	4.39	1.26	4.30	1.37	4.75***	1.09	3.61***	1.29

Note: *p < .05, **p < .01, ***p < .001.

Table 2
Means, standard deviation and Pearson correlation matrix for continuous variables (N = 212)

	M	SD	1	2	3	4	5	6
1. Quantity of Ideas	12.46	6.46	-					
2. Originality of Ideas	3.17	.55	−.019	-				
3. Feasibility of Ideas	3.40	.63	.034	−.051	-			
4. Fun of Ideas	3.31	.66	.027	.37**	.33**	-		
5. Flexibility of Ideas	6.36	2.11	.75**	−.078	−.039	−.003	-	
6. Impression Management	4.34	1.28	−.045	.089	−.044	.036	−.050	(.75)

*p < .05, **p < .001.
Cronbach's alphas are shown in the diagonal.

Data availability

This study was pre-registered at Open Science Framework, and all data are available at:
https://osf.io/h6ztr/

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